

Department of CSE

TUTORIAL PLAN

Name of the Course-Coordinator: Prof. Kapil Sharma
Prof. Satinderpal Singh
Prof. Khushi
Prof. Preetkamal Singh

Branch	Semester	Subject	Code	Session
CSE	3 rd	Computer Architecture	CCS103	July-Dec.2025

Tutorial Sheet No. 3

Course-Content Covered:

Instruction codes - Direct and Indirect Address, Computer registers, Computer instructions, Timing and control, Instruction cycle, Instruction format, Memory, register, and input-output reference instructions, Input/ Output and interrupts, Design and working of a complete basic computer, Control functions, Design of accumulator logic.

Question 1) Explain the following:

- Explain the concept of timing and control in a computer system. Analyze why synchronization of operations is critical when multiple components (CPU, memory, and I/O devices) are working together.
- Analyze how interrupts affect the normal execution of instructions and evaluate the trade-offs between enabling and disabling interrupts.

Question 2) Explain the following:

- Draw the complete instruction cycle diagram where $D5 T3 : SC \leftarrow 0$.
- Why do some instructions (like BSA, ISZ) affect control flow, while others (like AND, LDA) do not?